

# MICHAEL WIECK-SOSA

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## EDUCATION

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**Carnegie Mellon University** | PhD in Statistics | Advisor: [Aaditya Ramdas](#) *Sept. 2022-Present*

- GPA: 3.98/4.00 | Awards: DeGroot-Goel Fellowship (2026), awarded by faculty to one Statistics PhD student per year
- Thesis topics: prediction with transformers, estimation of time series models, and inference for dependence relationships

**University of Illinois at Urbana-Champaign** | MS in Statistics *May 2022*

- GPA: 3.95/4.00 | Awards: two years of guaranteed assistantships with full tuition waiver and stipend

**Fordham University** | BS in Mathematics with Minors in Computer Science and Economics *May 2020*

- GPA: 3.77/4.00 | Awards: *magna cum laude* | GRE: 170/170 Quantitative, 163/170 Verbal, 4.5/6.0 Writing

## PROGRAMMING LANGUAGES AND TOOLS

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- Python expert: extensive experience with NumPy, pandas, Polars, scikit-learn, statsmodels, and PyTorch
  - R expert: extensive experience with dplyr, Rcpp, xts, zoo, caret, mgcv, glmnet, parallel, and ggplot2
  - SQL, Git, Bash expert: extensive experience with data management, version control, and Linux
  - C++ proficient: used for courses in Data Structures and Algorithms
  - q/kdb+ proficient: used during internship to work with high-frequency financial data

## COURSEWORK

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- **Statistics:** Advanced Statistical Theory, Intermediate Statistics, Mathematical Statistics, Advanced Time Series Analysis, Regression Analysis, Computational Statistics
  - **Computer Science:** Algorithms, Data Structures, Theory of Computation, Operating Systems, Computer Architecture, Artificial Intelligence, Machine Learning, Data Mining for Listening to the Social Universe
  - **Math:** Stochastic Calculus, Measure-theoretic Probability, Functional Analysis, Measure Theory, Interacting Particle Systems, Geometric Flows, Differential Geometry, Lie Groupoids & Lie Algebroids, Topology, Abstract Algebra, Numerical Analysis, Numerical Linear Algebra, Real Analysis, Differential Equations, Linear Algebra, Mathematical Modeling

## DOCTORAL RESEARCH

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**Invariance-based Post-training of Transformers for Time Series Prediction** | Aaditya Ramdas

- Improving the quality of forecasts made by pre-trained transformers with a method inspired by invariant causal prediction

**Invariant Causal Prediction for Dependent Sequences in Nonstationary Environments** | Aaditya Ramdas

- Creating a statistical method that can be used to make machine learning models more robust to nonstationarity

**Dynamic Variable Importance for Time Series Foundation Models** | Tom Zhang and Chad Schafer

- Quantifying how predictor importance changes across lags and forecast horizons in models like Chronos-2 and TabPFN-TS

**The Signature of a Large Language Model** | Leif Weatherby, Tyler Shoemaker, and Cosma Shalizi

- Designing methods for analyzing sequences output by large language models using advanced stochastic process theory

**Goodness-of-fit Testing and Confidence Sets through Random Features** | Cosma Shalizi

- Building a framework for uncertainty quantification and testing whether the law of a stochastic process is in a model class

**Estimating Dynamic Models by Matching Random Features** | Cosma Shalizi

- Developing a likelihood-free method for estimating the parameters of complicated models for time series

**Conditional Independence Testing for Nonstationary Time Series** | Michel F. C. Haddad and Aaditya Ramdas

- Creating a conditional independence test based on regression that is robust to nonstationarity and temporal dependence

## GRADUATE RESEARCH ASSISTANTSHIPS

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**Carnegie Mellon University** | [NSF Grant](#) | PI: Cosma Shalizi *May 2024-Present*

- Developing theory, methods, and software in Python and R for estimating and inferring the parameters of dynamic models
- Using optimization algorithms to optimize highly non-convex objective functions to fit complicated models for time series
- Applied methods to time series models, state-space models, SDEs, ODEs, and dynamical systems with observational noise

**University of Illinois at Urbana-Champaign** | FORWARD Data Lab | Computer Science Department *Jan. 2021-May 2021*

- Discovered patterns in cross-platform dynamics on Twitter, Facebook, and Reddit with Hawkes processes using Python

**National Center for Supercomputing Applications** | Innovative Software and Data Analysis Group *Sept. 2020-May 2022*

- Built confidence bands for trends in concentrations and fluxes of chemicals to measure changes in water quality over time
- Used parallel computing to construct these confidence bands for 1000+ locations in the U.S. and made visualizations using R

## INDUSTRY INTERNSHIPS

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**J.P. Morgan** | Quantitative Research | Markets Summer Associate | Equity Derivatives and Macro *June 2023-Aug. 2023*

- Worked with macro traders and quants on a hedging method for derivatives portfolios via multi-period optimization
- Collaborated with energy derivatives traders on improving the statistical techniques used in a new trading strategy

**J.P. Morgan** | Quantitative Research | Markets Summer Associate | Equity Derivatives *June 2022-Aug. 2022*

- Developed method for adaptively selecting parameters of trade execution algorithms based on real-time data
- Built pipeline for processing market microstructure data using q/kdb+ and making predictions using scikit-learn in Python
- Used Bayesian optimization and walk-forward cross-validation to tune the hyperparameters of machine learning models

## RESEARCH INTERNSHIPS

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**MIT Lincoln Lab** | Group 38 *May 2021-July 2021*

- Implemented optical tracking methods for tracking objects in outer space and ran simulations to compare approaches

**Corteva Agriscience** | Research & Development Division *June 2020-Aug. 2020*

- Analyzed environmental changes based on time series of 50+ weather and soil features from 1,000+ sensor sites in the U.S.
- Used spatiotemporal Gaussian mixture models, dimensionality reduction, and clustering techniques with Python
- Built an interactive Dash application to visualize the statistical analysis of how environments changed over space and time

**Fordham University** | Robotics and Computer Vision Lab | Computer Science Department *Aug. 2019-March 2020*

- Created a real-time tracking system for the Kihansi spray toad in collaboration with zoologists using deep learning in Python

## TEACHING ASSISTANTSHIPS

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- Courses in the MS in Computational Finance program at CMU: Simulation Methods for Option Pricing, Financial Time Series, Financial Data Science I and II, and the Machine Learning Capstone Project (x2)
  - Courses in the MS in Data Science program at CMU: Time Series and Experimental Design
  - Courses in the BS in Statistics and Statistics/Machine Learning programs at CMU: Advanced Data Analysis

## PROFESSIONAL SERVICE

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- Reviewer (1 paper) | Journal of the Royal Statistical Society, Series B (Statistical Methodology) *2026*
  - Chair | Statistics PhD Student Committee for Organizing Career Events *2024-2025*

## TALKS AND POSTERS

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- In-person Talk (20 min.) | Conditional Independence Testing | International Workshop in Sequential Methodologies *2026*
  - In-person Talk (1 hour) | Simulation-based Inference | Statistical Methods for the Physical Sciences Group at CMU *2025*
  - Poster | Simulation-based Inference | Workshop on Neural Simulation-based Inference at CMU *2025*
  - Online Talk (1 hour) | Variable Selection for Forecasting | Presentation to Research Scientists at Amazon *2025*
  - Online Talk (20 min.) | Conditional Independence Testing | Virtual Time Series Seminar Junior Researcher Workshop *2025*
  - Poster | Conditional Independence Testing | NBER-NSF Time Series Conference at the University of Pennsylvania *2024*
  - In-person Talk (1 hour) | Nonlinear Regression for Time Series | Statistics and Machine Learning Group at CMU *2024*